

Heal the Planet

Complete Game Design Document

Earth Month Card Game Prototype

Field	Details
Working Title	Heal the Planet
Genre	Environmental strategy card game; solo or cooperative crisis management
Format	Deck-based tactical card play with randomized crises, readiness-index scoring, and a final global crisis
Session Use	30-minute guided prototype demo and audience brainstorm
Player Count	1 solo player, 2-4 cooperative players, or one audience group acting as a Global Council
Document Note	The game-facing design avoids direct references to specific real-world events or movements. A real-world inspiration table is retained in the appendix for internal reference.

1. High Concept

Heal the Planet is a card-based environmental strategy game where players work to protect Earth from escalating ecological crises. Over 10 turns, smaller planetary emergencies arise that damage Planet Health, reduce public confidence, strain resources, weaken ecosystems, or lower global readiness. Players respond by playing cards themed around education, community action, environmental restoration, policy work, emergency response, and green technology.

The goal is to keep the planet stable long enough to face a Final Global Crisis at the end of Turn 10. Victory depends not only on surviving immediate damage, but also on building the readiness indexes needed to withstand the final threat.

2. Design Pillars

- **Hope Through Action:** The game should show that coordinated action can meaningfully improve planetary outcomes.
- **Tradeoffs, Not Lectures:** Players should learn through choices, consequences, opportunity costs, and imperfect solutions.
- **Many Solutions Working Together:** Education, ecological repair, policy, economic transition, technology, and emergency response all matter.
- **Escalating Pressure:** Each turn should feel more urgent as the Final Global Crisis approaches.

- **Deck Identity:** The game should feel like a card game, not just a score tracker. Drawing, discarding, upgrading, exhausting, and managing status cards should matter.
- **Session-Friendly Play:** The prototype should be easy to demonstrate, vote on, and brainstorm around in a short workshop.

3. Target Experience

Players should feel tense but hopeful. Each turn asks them to choose between immediate relief and long-term preparedness. A good turn should create discussion: Should the group stop this crisis now, build public trust for future policy, invest in technology, repair damaged ecosystems, or accept short-term damage to prepare for a stronger future turn?

The game should avoid simple “right answer” choices. A strong card may raise one readiness index while lowering another, or solve an immediate problem while creating a long-term cost.

4. Player Count and Format

- **Solo version:** 1 player manages the planet using a shared deck.
- **Cooperative version:** 2-4 players discuss and play from a shared hand or individual hands.
- **Workshop version:** The audience acts as a Global Council, voting on which cards to play each turn.

5. Core Game Loop

- **Reveal Crisis Card:** A smaller planetary crisis occurs and applies immediate or delayed harm.
- **Draw Action Hand:** Players draw 5 cards from the Action Deck.
- **Gain Resources:** Players receive 3 Action Points, plus any ongoing bonuses.
- **Play Action Cards:** Players spend resources to mitigate damage, raise readiness indexes, remove status cards, or prepare long-term defenses.
- **Resolve Effects:** Apply mitigation, ongoing effects, status cards, crisis consequences, and index changes.
- **Check Planet Status:** If Planet Health reaches 0, the players lose.
- **Advance Turn Counter:** After Turn 10, reveal and resolve the Final Global Crisis.

6. Core Resources and Components

Resource / Component	Purpose	Prototype Rule
Action Points	Main cost paid to play cards each turn.	Players gain 3 Action Points per turn.
Policy Points	Used to approve high-impact Policy Cards.	Gained through education, organizing, research, and governance cards.
Planet Health	Overall survival score of the planet.	Starts at 20 in the short prototype; can be higher in the full version.
Readiness Indexes	Four long-term scores that determine final crisis resilience.	Each starts at 3 and can rise to 10.
Status Cards	Negative cards that clog the deck or reduce efficiency.	Added by crises such as Pollution, Apathy, Delay, and Misinformation.
Project Aids	Advisor cards chosen before play to shape strategy.	Choose up to 3 at setup.
Crisis Deck	Randomized smaller environmental problems.	Reveal 1 Crisis Card each turn.

Resource / Component	Purpose	Prototype Rule
Final Crisis Card	The major endgame test after Turn 10.	Resolved after the 10th turn.

7. Planetary Readiness Indexes

The game uses four readiness indexes from 0 to 10. These indexes determine whether the planet can survive the Final Global Crisis. At the start of the game, each index begins at 3, meaning the world is vulnerable but recoverable.

Index	Represents	Gameplay Role
Public Trust	Education, awareness, cooperation, and confidence in environmental action.	Prevents apathy, misinformation, and resistance to change.
Ecological Stability	Biodiversity, clean water, soil, forests, and ecosystem resilience.	Reduces damage from pollution, habitat loss, and extreme environmental events.
Green Economy	Sustainable jobs, clean infrastructure, funding, and technological transition.	Makes powerful solutions more affordable and keeps recovery economically viable.
Global Coordination	Policy strength, emergency planning, treaties, and institutional capacity.	Prevents calamities and reduces large-scale crisis damage.

Index Scale

- 0-2: Critical
- 3-5: Vulnerable
- 6-8: Stable
- 9-10: Resilient

8. Card Types

Cards should create tactical choices and long-term consequences. A card can raise one index while lowering another, heal Planet Health, remove status cards, generate Policy Points, preview crises, or create ongoing protection.

Card Type	Role	Common Effects
Education	Builds public understanding and cooperation.	Raises Public Trust, removes Apathy or Misinformation, draws cards, generates Policy Points.
Environmental Work	Directly restores ecosystems and reduces pollution.	Raises Ecological Stability, heals Planet Health, removes Pollution, prevents damage from land, water, or wildlife crises.
Technology	Creates powerful long-term systems and infrastructure.	Raises Green Economy, creates ongoing prevention, previews crises, or makes future solutions cheaper.
Policy	Represents institutional, regional, or global action.	Costs Policy Points, raises Global Coordination, creates lasting rules, prevents calamities.
Emergency Response	Handles immediate disaster impact.	Prevents incoming damage, restores Planet Health, protects indexes, buys time.

9. Deck Behavior and Card Progression

This section preserves the stronger deckbuilding identity from the earlier concept. The readiness indexes give the game strategic direction, while the deck system gives each turn tactical texture.

Basic Deck Rules

- Players start with a small starter deck of 10 cards.
- At the start of each turn, draw 5 cards.
- Players have 3 Action Points to spend each turn.
- Played cards go to the discard pile unless they have the Exhaust keyword.
- Unplayed cards are discarded at the end of the turn unless a card effect says otherwise.
- When the deck runs out, shuffle the discard pile into a new deck.
- Status Cards can be added to the deck by crises and must be removed through card effects.

Card Keywords

Keyword	Meaning	Design Use
Draw	Draw additional cards.	Creates combo turns and lets Education or Research cards support planning.
Exhaust	Remove this card from the deck after it is played for the rest of the game.	Used for powerful one-time emergency actions or major policies.
Ongoing	The card remains active and provides a benefit each turn.	Used for technology, infrastructure, and long-term policy cards.
Upgrade	Improve a card into a stronger version.	Rewards planning, research, or investment actions.
Retain	Keep this card in hand instead of discarding it at the end of the turn.	Useful for expensive cards that require setup.
Cleanse	Remove a Status Card from the hand, discard pile, or deck.	Gives players a way to recover from pollution, apathy, or delay.

Status Cards

Status Card	Effect	Removed By
Pollution	When drawn, lose 1 Action Point unless discarded by a cleanup effect.	Environmental Work or Cleanse effects.
Apathy	Cannot be played. Takes up space in the hand.	Education cards or Public Trust effects.
Misinformation	The next Education or Policy card costs 1 more.	Education cards, Public Trust effects, or specific response cards.
Delay	The next Technology or Policy card played this turn has no bonus effect.	Global Coordination or Planning effects.
Backlash	Lose 1 Public Trust when this card is drawn, then Exhaust it.	Public Trust or Community Response effects.

10. Example Action Cards

Card	Type	Cost	Effect
Community Workshop	Education	1 AP	+2 Public Trust. Remove 1 Apathy or Misinformation status card.
School Green Program	Education	1 AP	+1 Public Trust. Draw 1 card.
Youth Organizer Network	Education	2 AP	+3 Public Trust. Gain 1 Policy Point.
Research Briefing	Education	1 AP	Draw 1 card. Gain 1 Policy Point.
Tree Canopy Program	Environmental Work	2 AP	+2 Ecological Stability. Restore 2 Planet Health.

Card	Type	Cost	Effect
River Cleanup	Environmental Work	1 AP	+1 Ecological Stability. Cleanse 1 Pollution card.
Community Compost Hub	Environmental Work	1 AP	Cleanse 1 Pollution. +1 Public Trust.
Habitat Corridor Project	Environmental Work	2 AP	+3 Ecological Stability. Future habitat crises deal -1 damage.
Renewable Grid Buildout	Technology	3 AP	+3 Green Economy, +1 Ecological Stability. Ongoing: prevent 1 climate-related damage each turn.
Water Purification Network	Technology	2 AP	+2 Ecological Stability. Prevent the next water crisis from dealing 2 damage.
Predictive Climate Modeling	Technology	2 AP	Look at the next 2 Crisis Cards. Put one on the bottom of the deck.
Clean Infrastructure Act	Policy	2 PP	+2 Global Coordination, +2 Green Economy.
Protected Habitat Law	Policy	2 PP	+3 Ecological Stability, +1 Global Coordination.
Emergency Response Network	Emergency	1 AP	Prevent 3 incoming crisis damage. +1 Public Trust if damage is fully prevented.

11. Example Starter Deck

A simple 10-card starter deck should give players a mix of education, restoration, emergency defense, and early infrastructure options.

#	Card	Purpose
1	Community Workshop	Build Public Trust and remove negative public-response cards.
2	Community Workshop	Repeatable early support.
3	River Cleanup	Restore Ecological Stability and remove Pollution.
4	River Cleanup	Repeatable environmental repair.
5	Tree Canopy Program	Restore ecosystems and Planet Health.
6	Community Compost Hub	Low-cost waste mitigation and Public Trust gain.
7	Emergency Response Network	Prevent immediate crisis damage.
8	Repair and Reuse Campaign	Reduce waste and generate Public Trust.
9	Solar Installation Project	Early Green Economy and technology setup.
10	Local Policy Petition	Generate Policy Points for future Policy Cards.

12. Crisis Cards

At the start of each turn, reveal one Crisis Card. Crisis Cards create immediate pressure, damage indexes, reduce Planet Health, add status cards, or trigger Calamity checks.

Crisis Card	Effect	Calamity Check
Heat Dome	Deal 2 Planet Health damage. -1 Ecological Stability.	If Global Coordination is 2 or lower, deal +2 damage.
Aging Water Infrastructure	-2 Public Trust, -2 Ecological Stability.	If Public Trust is 2 or lower, add 1 Apathy card.
Industrial Pollution Corridor	Add 2 Pollution status cards. -1 Ecological Stability.	If Ecological Stability is 2 or lower, trigger Ecosystem Collapse.

Crisis Card	Effect	Calamity Check
Plastic Waste Surge	Add 1 Pollution card. Deal 2 Planet Health damage.	If no Environmental Work card is played this turn, add another Pollution card.
Public Burnout	-2 Public Trust. Draw 1 fewer card next turn.	If Public Trust is 2 or lower, trigger Mass Apathy.
Deforestation Surge	-3 Ecological Stability.	If untreated this turn, lose 2 additional Planet Health.
Supply Chain Shock	-2 Green Economy. Technology cards cost +1 this turn.	If Green Economy is 2 or lower, trigger Economic Pushback.
Diplomatic Gridlock	-2 Global Coordination. Policy cards cost +1 Policy Point this turn.	If Global Coordination is 2 or lower, trigger Policy Paralysis.

13. Calamities

Calamities are major negative events caused by low readiness indexes. They are not drawn every turn by default. Instead, Crisis Cards check whether an index has fallen into the Critical range.

Calamity	Trigger	Effect
Mass Apathy	Public Trust is 2 or lower.	Draw 1 fewer card next turn and add 1 Apathy status card.
Ecosystem Collapse	Ecological Stability is 2 or lower.	Lose 4 Planet Health and add 1 Pollution status card.
Economic Pushback	Green Economy is 2 or lower.	Technology and Policy cards cost +1 next turn.
Policy Paralysis	Global Coordination is 2 or lower.	Players cannot play Policy Cards next turn.
Cascading Disaster	Two or more indexes are Critical.	Lose 5 Planet Health and add 1 Apathy status card.

14. Project Aids

Before the game begins, players choose up to three Project Aids. These are fictional advisors, specialists, or organizations that shape the player's strategy. Each Project Aid has a passive bonus and a limitation.

Project Aid	Role	Passive Ability	Drawback
The Educator	Community learning specialist	First Education card each turn gives +1 additional Public Trust.	Technology cards cost +1 if Public Trust is below 4.
The Ecologist	Habitat and wildlife specialist	Once per turn, when Ecological Stability is restored, restore +1 additional point.	Green Economy starts 1 point lower.
The Engineer	Clean systems designer	First Technology card each turn costs 1 less.	Public Trust gains are reduced by 1 if no Education card was played this turn.
The Policy Advocate	Institutional strategist	Gain +1 Policy Point whenever a Policy Card is played.	If Global Coordination drops below 3, lose 1 Public Trust.
The Disaster Responder	Emergency planner	First crisis damage each turn is reduced by 1.	Long-term solution cards do not gain bonus effects.

15. Final Global Crisis

After Turn 10, reveal the Final Global Crisis. The Final Crisis checks all four readiness indexes. The stronger the indexes, the more damage is prevented. If the world ignored one area for too long, the final event becomes much harder to survive.

Recommended Final Crisis: Cascading Planetary Crisis

- Base Effect: Deal 20 damage to Planet Health.
- Public Trust 6 or higher: Reduce damage by 3.
- Ecological Stability 6 or higher: Reduce damage by 4.
- Green Economy 6 or higher: Reduce damage by 3.
- Global Coordination 6 or higher: Reduce damage by 4.
- If all four indexes are 6 or higher, reduce damage by an additional 3.
- If any index is 2 or lower, increase damage by 3.
- If two or more indexes are 2 or lower, trigger Cascading Disaster before applying final damage.

16. Win, Loss, and Final Rating

- Win: Planet Health remains above 0 after the Final Global Crisis resolves.
- Loss: Planet Health reaches 0 at any point.
- Optional loss condition: Public Trust reaches 0, causing global inaction.

Rating	Condition
Collapse	Planet Health reaches 0.
Survival	Planet Health is above 0, but two or more indexes remain Critical or Vulnerable.
Stabilization	Planet Health is above 0, and at least two indexes are Stable.
Resilient Future	Planet Health is above 0, and all four indexes are Stable or better.
Regenerative Future	Planet Health is above 10, and all four indexes are Resilient.

17. Example Turn

Turn 4 begins. The players reveal a Crisis Card: Industrial Pollution Corridor. It adds 2 Pollution status cards and lowers Ecological Stability by 1. If Ecological Stability is 2 or lower, it will trigger Ecosystem Collapse.

The players draw 5 cards: River Cleanup, Community Workshop, Renewable Grid Buildout, Emergency Response Network, and Tree Canopy Program. They have 3 Action Points.

- Play River Cleanup for 1 AP: +1 Ecological Stability and Cleanse 1 Pollution card.
- Play Emergency Response Network for 1 AP: prevent 3 incoming crisis damage if any is applied this turn.
- Play Community Workshop for 1 AP: +2 Public Trust and remove 1 Apathy or Misinformation card if present.
- They cannot afford Renewable Grid Buildout this turn, so they delay a powerful long-term technology card.

Result: The immediate crisis is mostly controlled, but the players miss a chance to build long-term Green Economy. This creates the central tension of the game: survive now or prepare for later.

18. Short Workshop Prototype

For a 30-minute Earth Month session, run a shortened version of the game. The goal is to demonstrate the core loop and then invite the audience to brainstorm new cards, crises, and Project Aids.

- Planet Health starts at 20.
- Each readiness index starts at 3.
- Players draw 5 cards each turn.
- Players gain 3 Action Points each turn.
- Play 3 sample turns, then jump to the Final Global Crisis.
- The audience acts as the Global Council and votes on which cards to play.

19. Example Workshop Flow

- 0-3 minutes: Introduce the premise and explain Planet Health, Action Points, and the four readiness indexes.
- 3-8 minutes: Show the starting hand, first Crisis Card, and Project Aids.
- 8-20 minutes: Play three compressed turns with audience voting.
- 20-25 minutes: Resolve the Final Global Crisis and reveal the final rating.
- 25-30 minutes: Ask the audience to invent new cards, crises, and Project Aids.

20. Prototype Components

- 20-30 Action Cards
- 10 Crisis Cards
- 1 Final Global Crisis Card
- 5 Project Aid cards
- Planet Health tracker
- Four readiness index trackers
- Action Point tokens
- Policy Point tokens
- Status cards: Pollution, Apathy, Delay, Misinformation, Backlash
- Turn tracker from 1 to 10

21. Playtest Questions

- Did players understand what each index represented?
- Did crisis cards create urgency without feeling unfair?
- Did cards create meaningful tradeoffs?
- Did the deck feel interesting to manage?
- Were Status Cards annoying in a productive way or just frustrating?

- Was the Final Global Crisis clearly connected to earlier decisions?
- Did the environmental theme come through naturally?
- Which card felt most satisfying to play?
- Which crisis felt most dramatic?
- What new action could become a card?
- Would players want to replay with different Project Aids?
- Was the short prototype understandable within 30 minutes?

Appendix A: Best Real-Life Inspirations Reference Table

This appendix is for internal design reference. The game-facing sections of the document avoid direct references to specific real-world events or movements, but the table below is retained as requested to preserve the research basis for adaptation.

Real-world basis	What it can inspire in the game	Possible card / mechanic
Earth Day / Earth Month	Mass public education and volunteer mobilization. Earth Day began in 1970 and grew into a global environmental movement. Source: earthday.org	Earth Month Mobilization: +2 Public Trust, gain 1 Policy Point, draw 1 card.
Fridays for Future / youth climate strikes	Youth-led public pressure, climate education, protest, and intergenerational urgency. Fridays for Future describes itself as a youth-led global climate strike movement that began in 2018. Source: fridaysforfuture.org	Youth Climate Strike: +3 Public Trust, but -1 Green Economy this turn due to disruption.
350.org fossil fuel divestment / clean energy activism	Campaigns to move society away from fossil fuels and toward clean energy. 350.org frames its work around ending fossil fuels and powering up clean, safe, affordable energy. Source: 350.org	Divestment Campaign: Gain 2 Policy Points. Future fossil-fuel crisis cards deal -1 damage.
Greenpeace-style direct action	Peaceful protest, public confrontation, and exposing environmental harm. Greenpeace describes its network as using peaceful protest and creative confrontation to expose environmental problems and promote solutions. Source: greenpeace.org	Expose Polluter: Cancel one corporate pollution effect, +1 Public Trust, add 1 Backlash status card.
Circular economy movement	Repair, reuse, refurbishment, recycling, composting, and designing waste out of systems. The Ellen MacArthur Foundation defines circular economy as keeping products and materials in circulation while regenerating nature. Source: ellenmacarthurfoundation.org	Circular Economy Shift: +2 Green Economy, +1 Ecological Stability. Waste crises deal -1 for the rest of the game.
Plastic reduction / plastic bans	Policies targeting single-use plastics and waste streams. U.S. Department of Energy materials note that increased plastic use and stagnant recycling rates drive concern about pollution and create a need for circular solutions. Source: energy.gov	Single-Use Plastic Ban: +2 Ecological Stability, +1 Global Coordination, -1 Green Economy.
Paris Agreement / global climate diplomacy	International cooperation and treaty-based climate action. The UN presents climate agreements and reports as part of global climate action efforts. Source: un.org	Global Climate Accord: Spend 3 Policy Points. +3 Global Coordination, +2 Public Trust.
Montreal Protocol / ozone recovery model	A hopeful example of global environmental cooperation. Use this as inspiration for a worldwide phaseout mechanic.	Global Phaseout Treaty: Choose one recurring crisis type. It deals -2 damage for the rest of the game.
Flint water crisis	Water infrastructure failure, environmental justice, public trust collapse, and data-driven repair. Columbia Climate School describes Flint's 2014 water switch and lead contamination from aging pipes. Source: news.climate.columbia.edu	Aging Water Infrastructure crisis: -3 Public Trust, -3 Ecological Stability unless repaired.
Environmental justice movement	Unequal pollution burdens and community-led resistance. NRDC describes environmental justice as focused on communities, especially communities of color, forced to live closest to pollution sources. Source: nrdc.org	Community Justice Coalition: +2 Public Trust, +2 Global Coordination. Prevent the next pollution calamity.
Cancer Alley / industrial pollution corridors	Long-term industrial pollution and unequal health burdens. A Princeton Legal Journal article describes Cancer Alley as an 85-mile stretch along the Mississippi River with significant petrochemical production and elevated cancer-risk concerns. Source: legaljournal.princeton.edu	Industrial Pollution Corridor crisis: Adds Pollution status cards every turn until addressed.
Biodiversity loss / IPBES global assessment	Species collapse, habitat destruction, and ecosystem services. IPBES provides scientific assessments on biodiversity and ecosystems for policymakers; reporting on its global assessment highlighted risks to up to 1 million species. Source: files.ipbes.net	Biodiversity Warning Report: Look at the next 2 Crisis Cards. Gain 1 Policy Point.

Real-world basis	What it can inspire in the game	Possible card / mechanic
Extreme weather intensification	Heatwaves, droughts, floods, storms, and cascading disasters. IPCC states climate change is intensifying the water cycle, increasing intense rainfall, flooding, and drought in many regions. Source: ipcc.ch	Extreme Weather Season crisis: Deal damage to two indexes; if Global Coordination is low, trigger a Calamity.
NOAA climate impact framing	Climate impacts as interconnected systems: food, health, infrastructure, disease, and productivity. NOAA notes drought can harm food production and health, while flooding can spread disease and damage ecosystems and infrastructure. Source: noaa.gov	Cascading Infrastructure Failure: Damages Green Economy and Ecological Stability together.
Renewable energy transition	Solar, wind, battery storage, grid modernization, and green jobs. Energy-transition research points to wind turbines, solar PV, lithium-ion batteries, EVs, and storage as important low-carbon technologies. Source: bakerinstitute.org	Renewable Grid Buildout: +3 Green Economy, +2 Ecological Stability, but requires Public Trust 5+.
Indigenous / Traditional Ecological Knowledge	Long-term stewardship, land knowledge, and justice-centered ecological planning. A policy paper discusses Traditional Ecological Knowledge and environmental justice issues such as uranium mining impacts on Navajo Nation lands. Source: pmc.ncbi.nlm.nih.gov	Traditional Ecological Knowledge: Remove 1 Calamity risk. +2 Ecological Stability, +1 Public Trust.